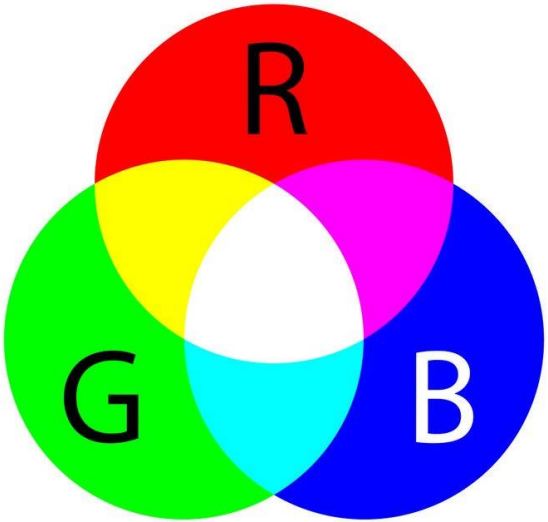
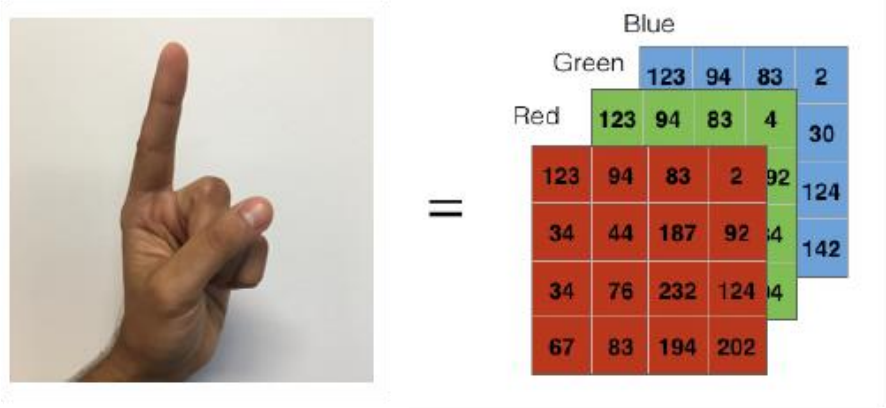
Deep learning avec Pytorch

Partie 7 : Computer vision

Présenté par **Morgan Gautherot**

Une image en informatique





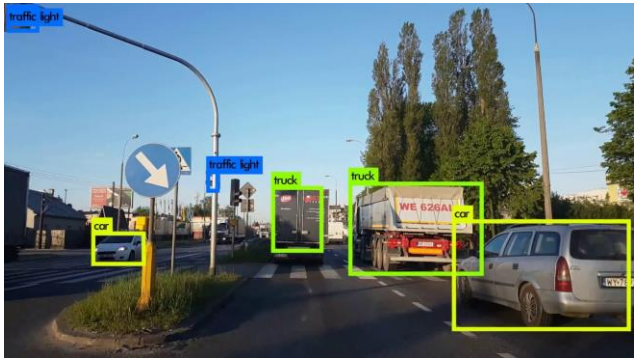
(4x4x3)

Computer vision

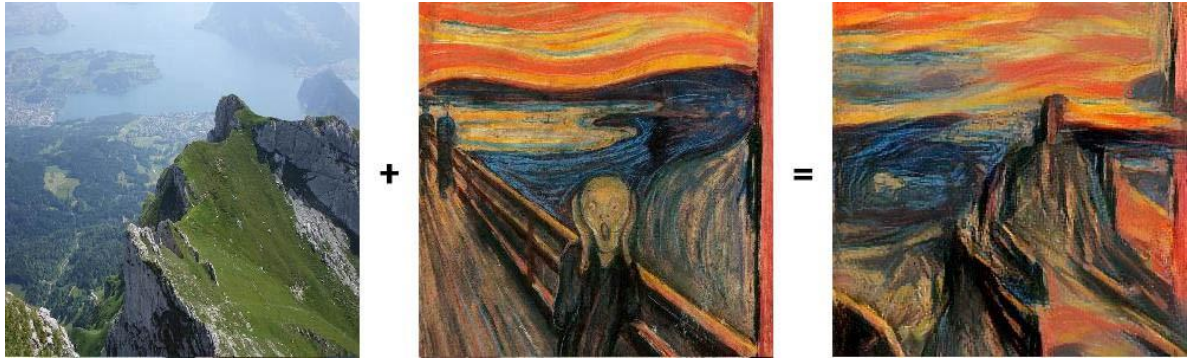
Classification d'images

Cat ? (0/1)

Détection d'objets



Transfert de style



Un grand nombre de caractéristiques

64 x 64

64 x 64 x 3

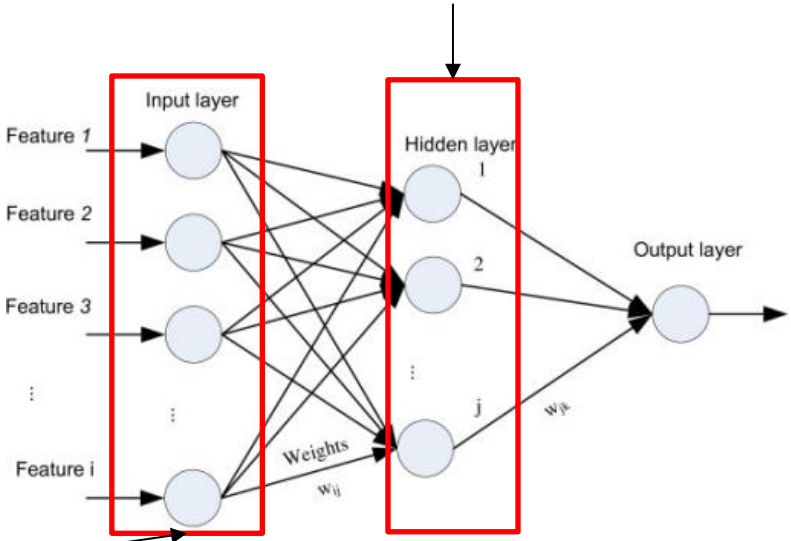
12288 caractéristiques

1000 x 1000

1000 x 1000 x 3

3 millions de caractéristiques

1 000 neurones

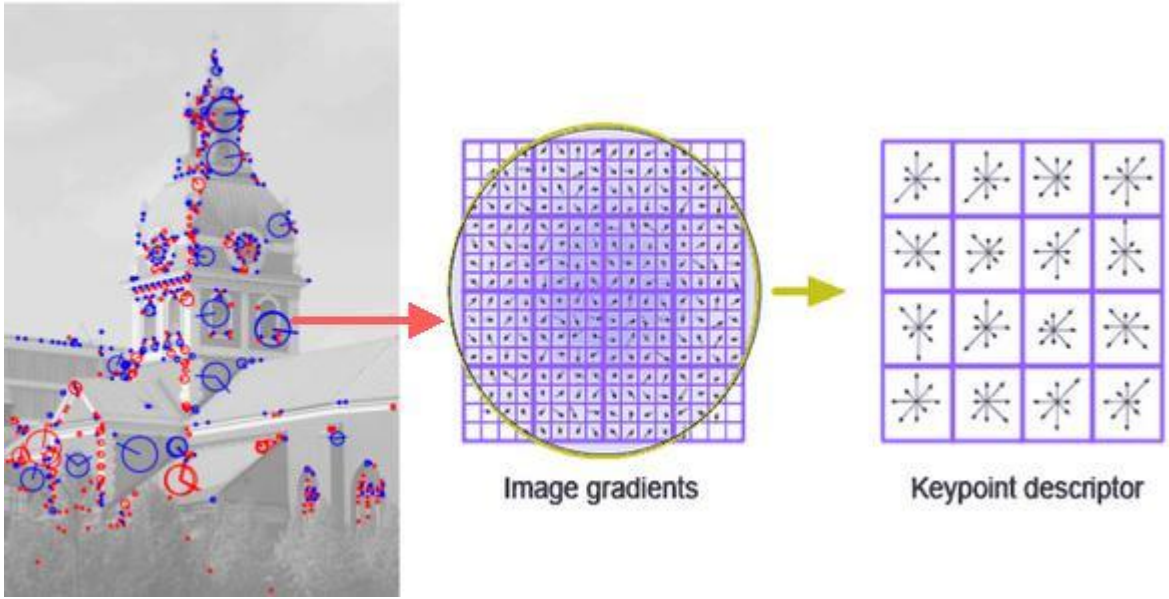


$W^{[1]} = 3$ milliards de paramètres

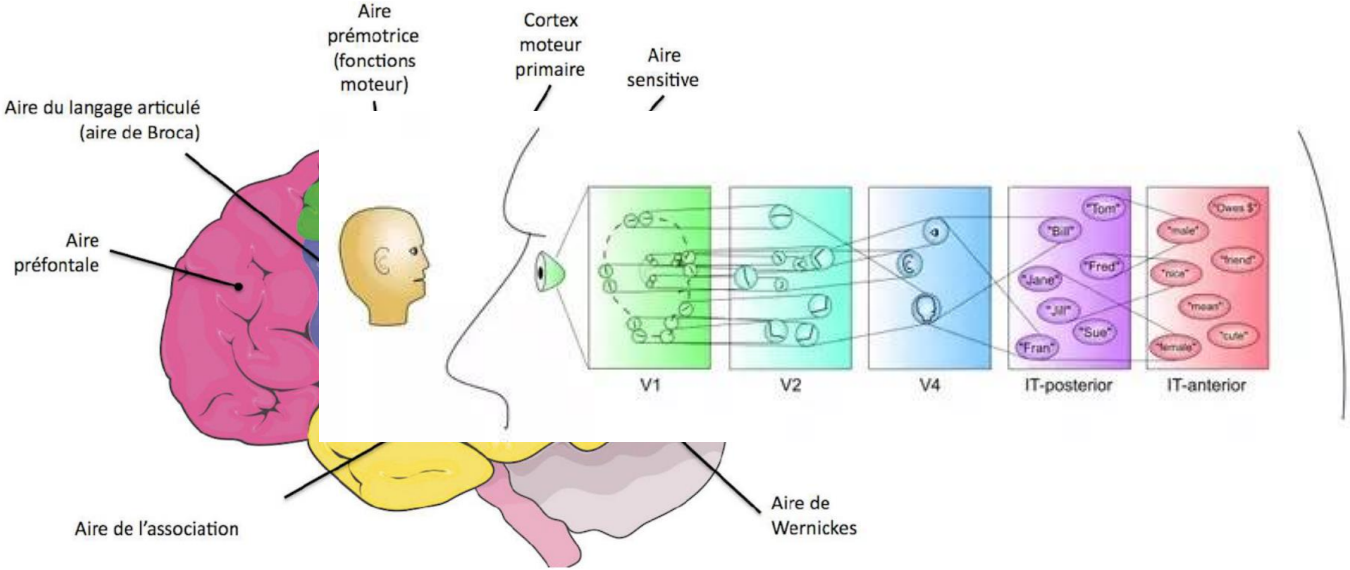
Extraction de caractéristiques

Partie 7 : Computer vision

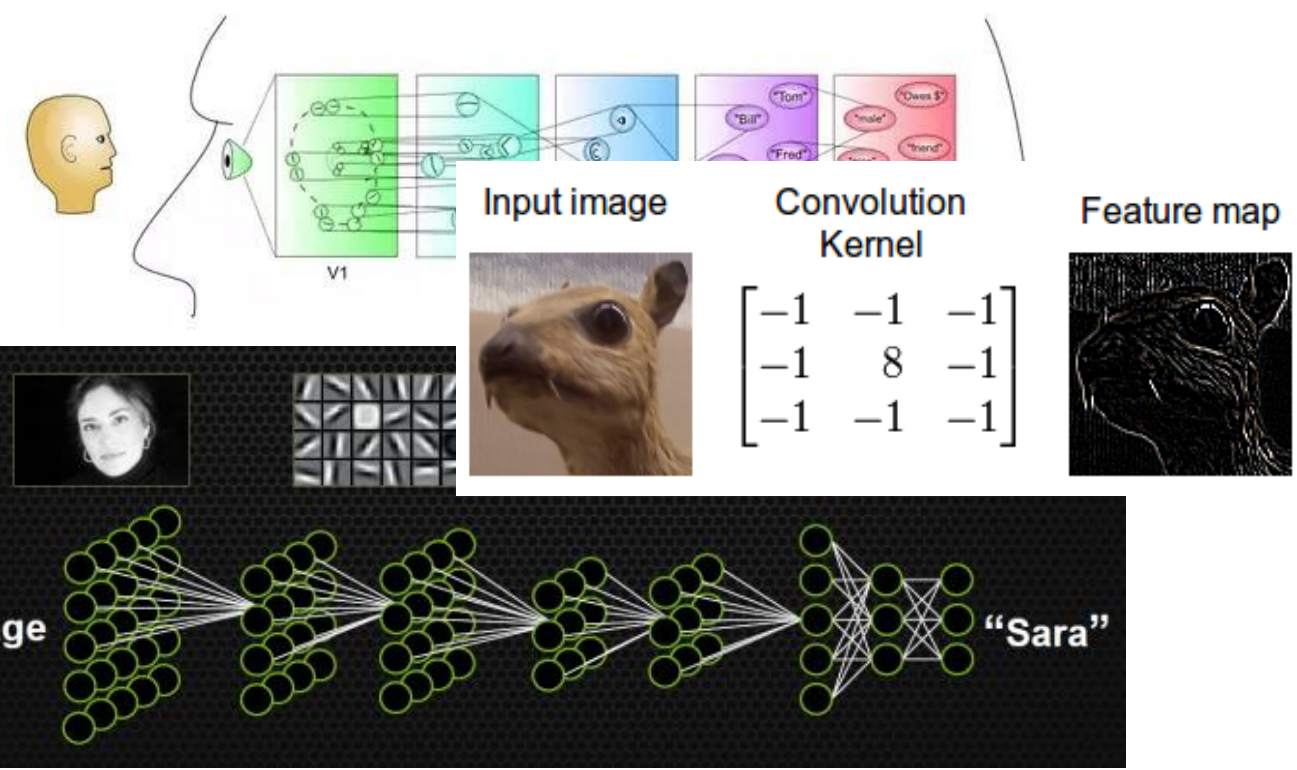
Extraction de caractéristiques



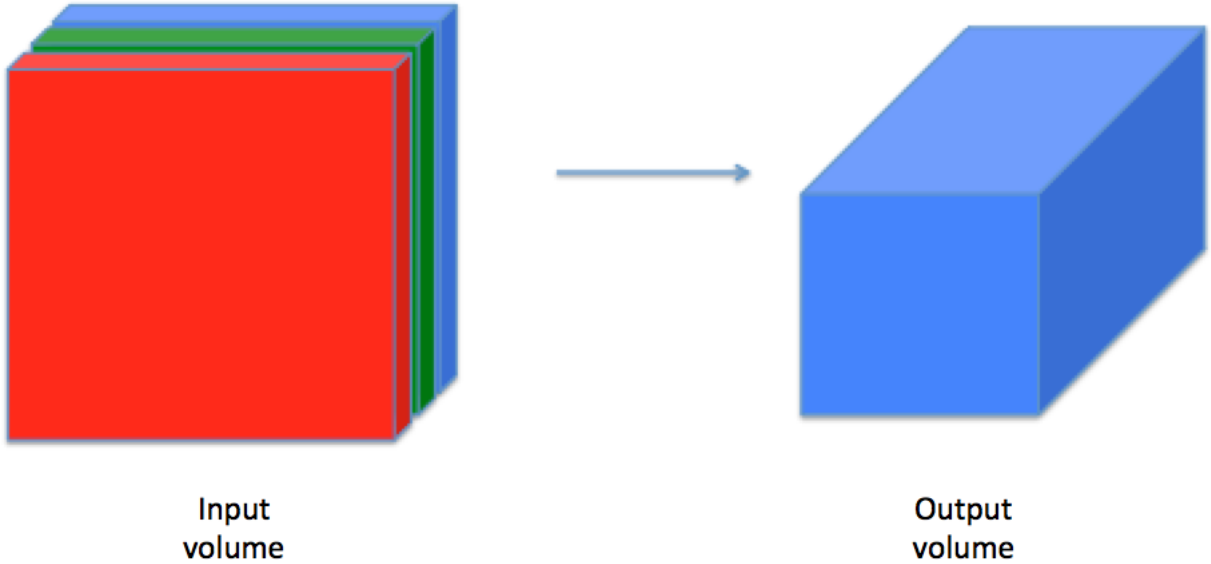
Comment voyons nous ?



De la biologie à l'informatique



Représentation vectorielle de l’image d’entrée



La convolution

Partie 7 : Computer vision

Les filtres

Figure 16.152. Edge detect

	0	1	0	
	1	-4	1	
	0	1	0	

Figure 16.153. Emboss

	-2	-1	0	
	-1	1	1	
	0	1	2	

Et mathématiquement ?

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

(6, 6)

Convolution

*

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

1	0	-1
1	0	-1
1	0	-1

(3, 3)

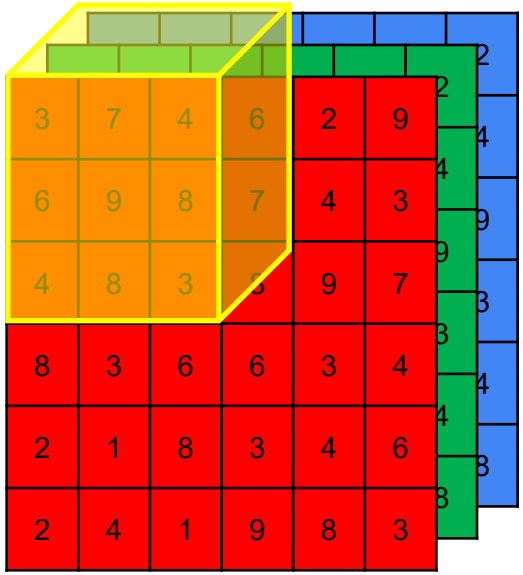
Filtre

=

-5	-4	0	8
-10	-2	2	3
0	-2	-4	-7
-3	-2	-3	-16

(4, 4)

Convolution pour les images RGB



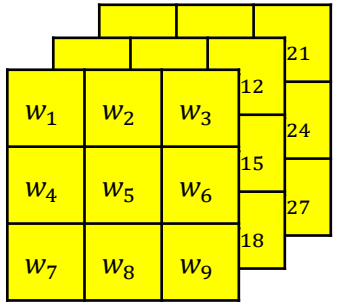
(6, 6, 3)

Hauteur

Largeur

channels

*



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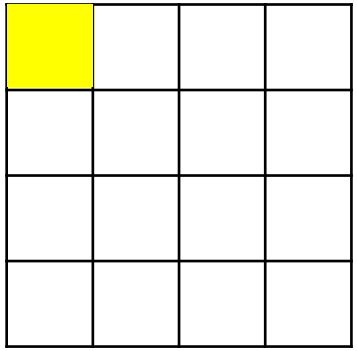


(3, 3, 3)

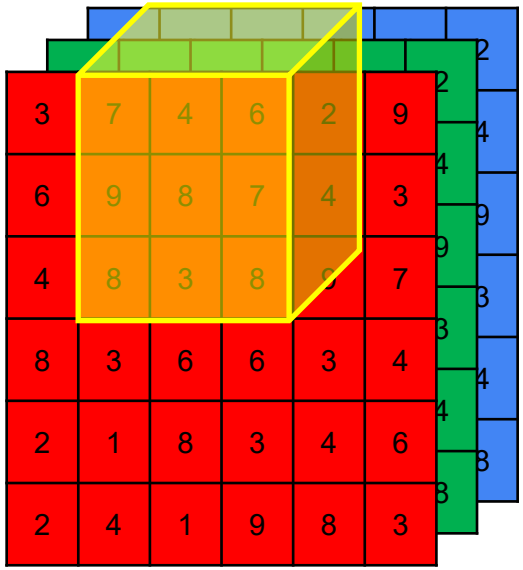
Hauteur

Largeur

channels

(4, 4, 1)

Convolution pour les images RGB



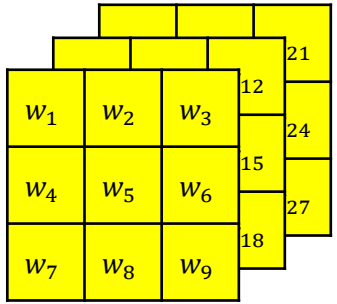
(6, 6, 3)

Hauteur

Largeur

channels

*



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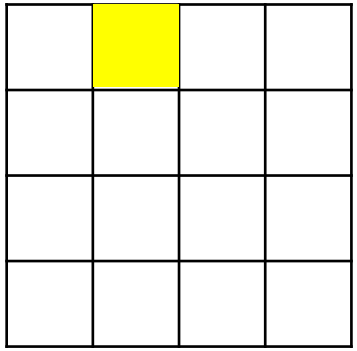


(4, 4, 1)

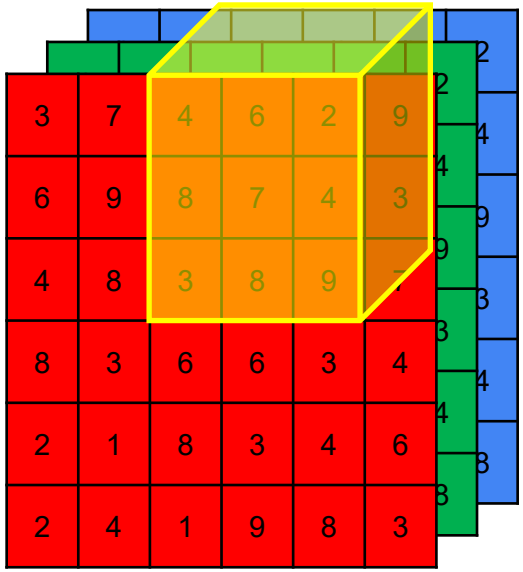
(3, 3, 3)

Hauteur

Largeur

channels

Convolution pour les images RGB



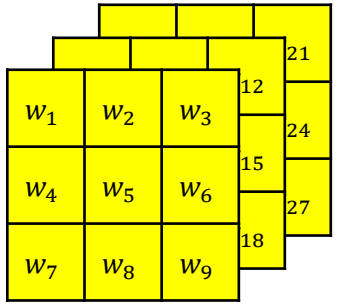
(6, 6, 3)

Hauteur

Largeur

channels

*



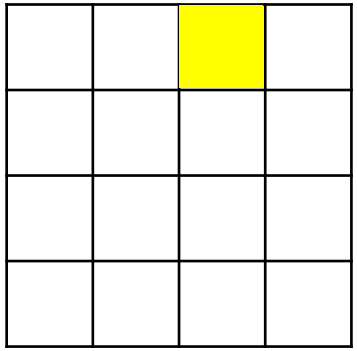
(3, 3, 3)

Hauteur

Largeur

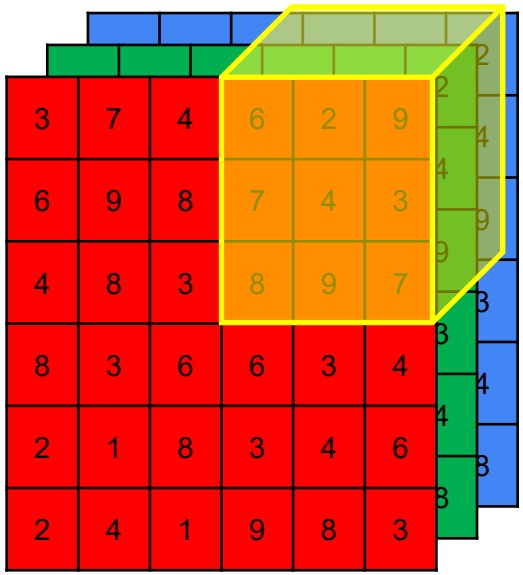
channels

=



(4, 4, 1)

Convolution pour les images RGB



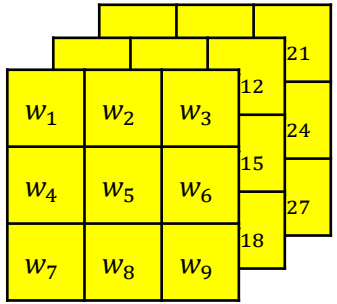
(6, 6, 3)

Hauteur

Largeur

channels

*



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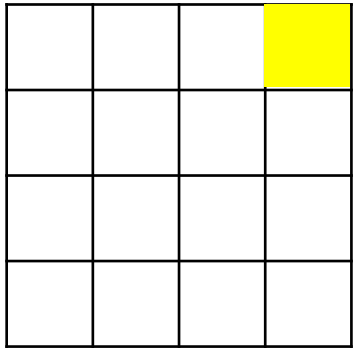


(3, 3, 3)

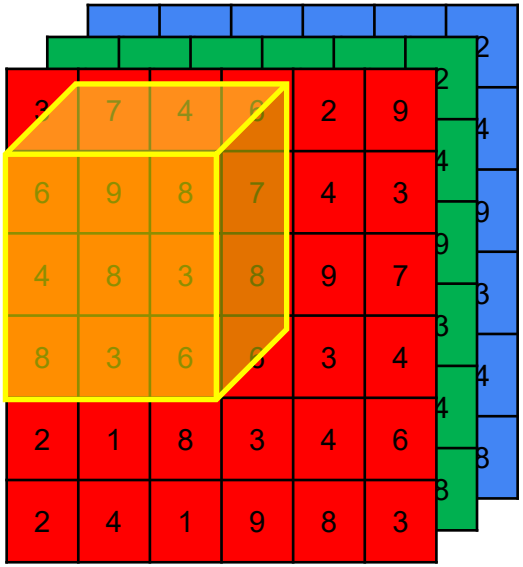
Hauteur

Largeur

channels

(4, 4, 1)

Convolution pour les images RGB



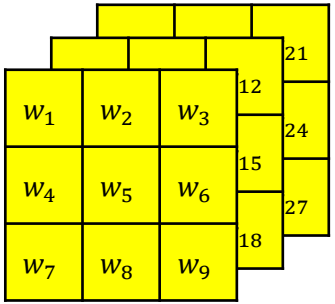
(6, 6, 3)

Hauteur

Largeur

channels

*



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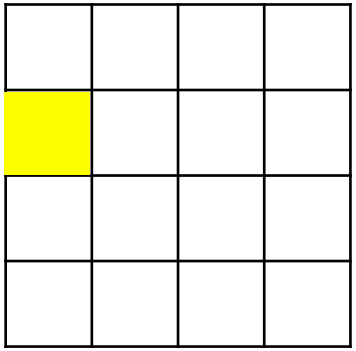


(3, 3, 3)

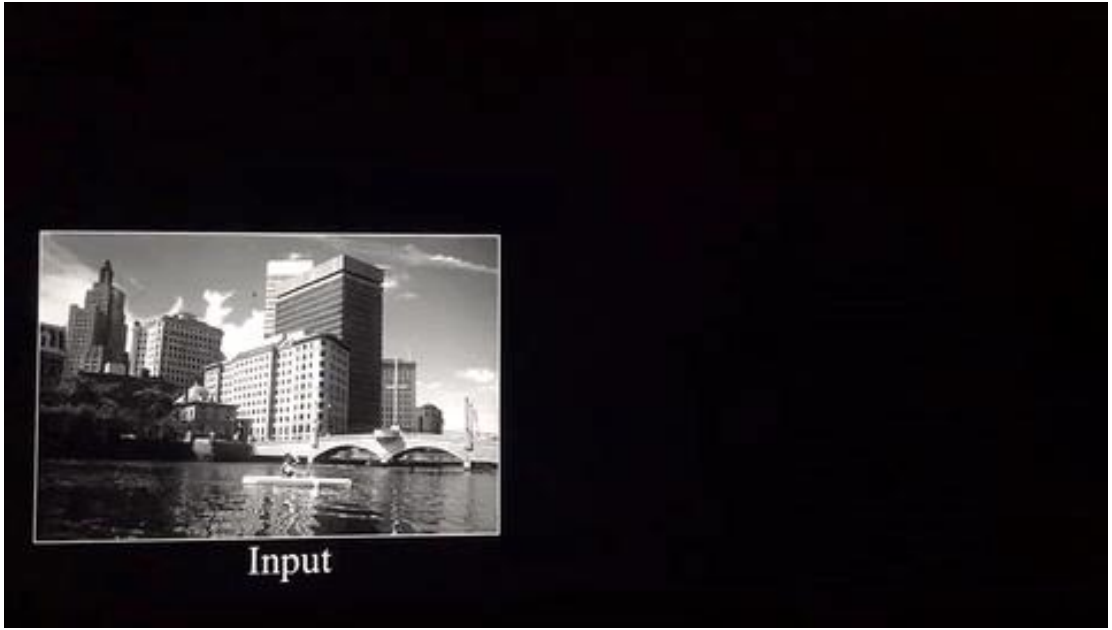
Hauteur

Largeur

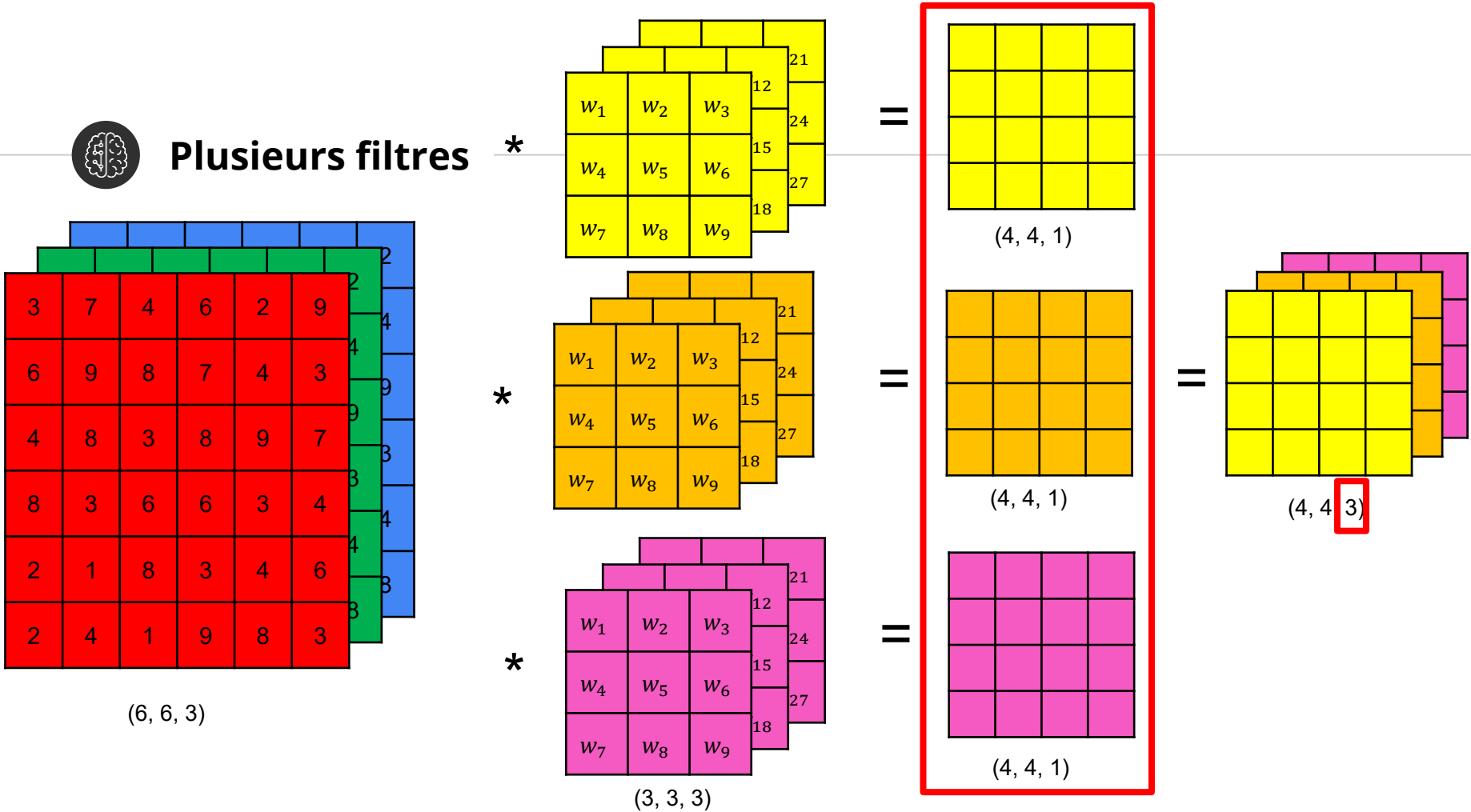
channels

(4, 4, 1)

Plusieurs filtres



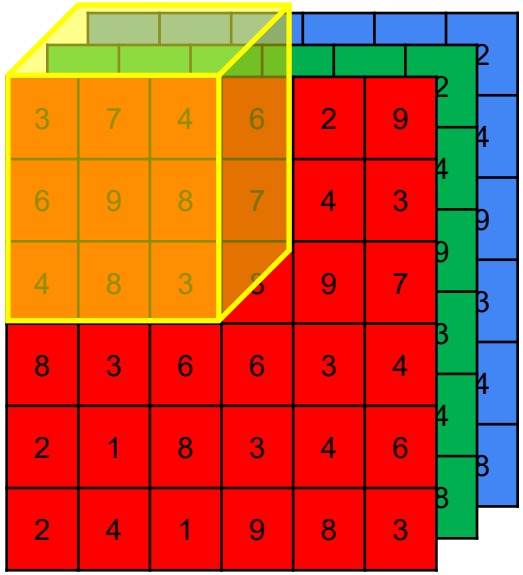
Plusieurs filtres



La convolution informatiquement

Partie 7 : Computer vision

Convolution pour les images RGB



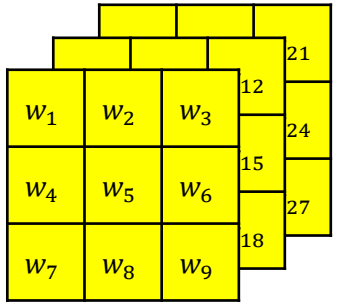
(6, 6, 3)

Hauteur

Largeur

channels

*



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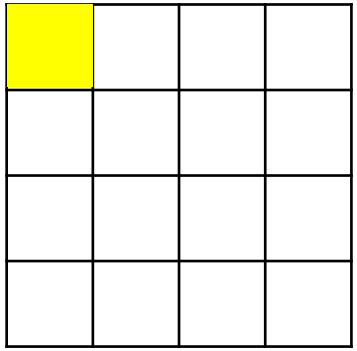


(3, 3, 3)

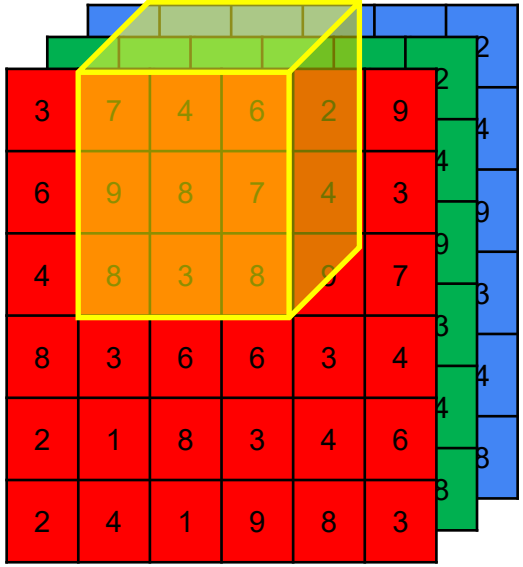
Hauteur

Largeur

channels

(4, 4, 1)

Convolution pour les images RGB



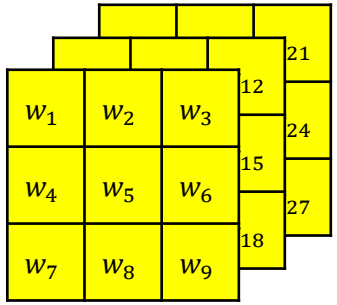
(6, 6, 3)

Hauteur

Largeur

channels

*



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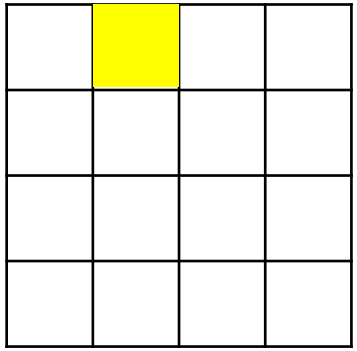


(3, 3, 3)

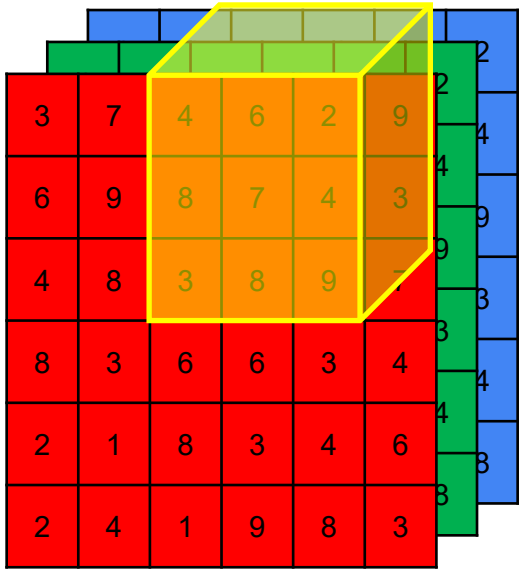
Hauteur

Largeur

channels

(4, 4, 1)

Convolution pour les images RGB



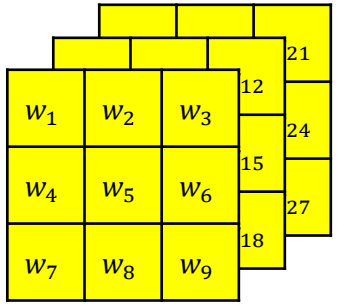
(6, 6, 3)

Hauteur

Largeur

channels

*



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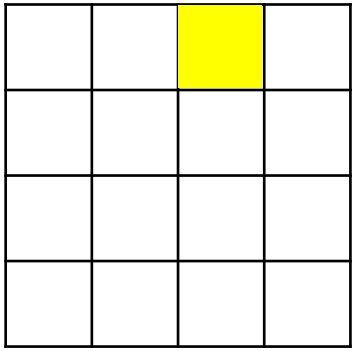


(4, 4, 1)

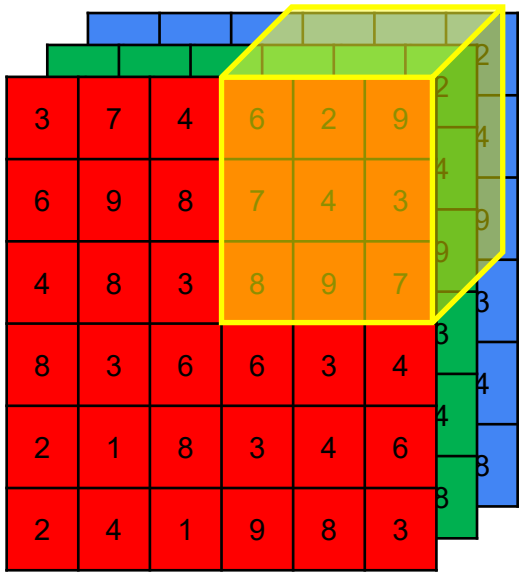
(3, 3, 3)

Hauteur

Largeur

channels

Convolution pour les images RGB



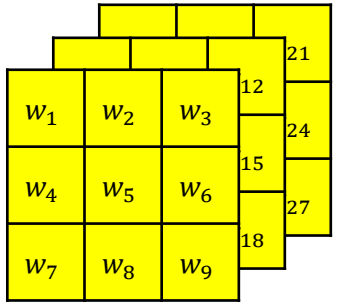
(6, 6, 3)

Hauteur

Largeur

channels

*



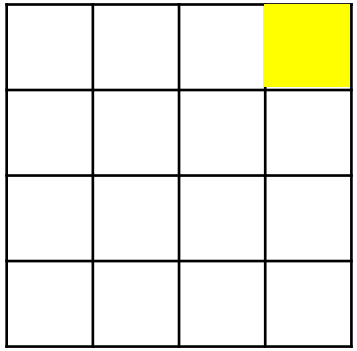
(3, 3, 3)

Hauteur

Largeur

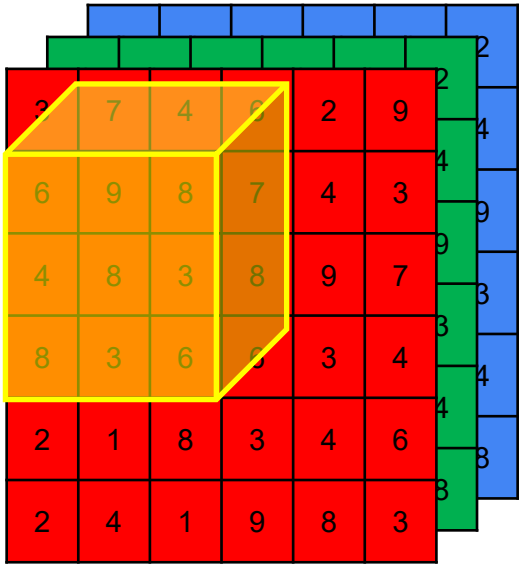
channels

=



(4, 4, 1)

Convolution pour les images RGB



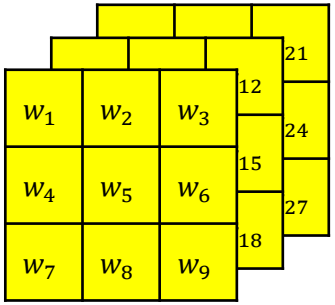
(6, 6, 3)

Hauteur

Largeur

channels

*



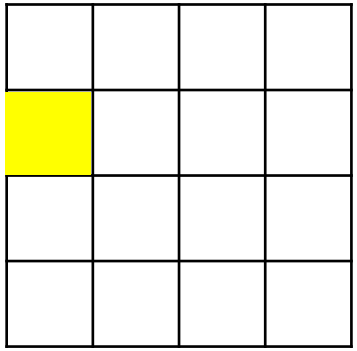
(3, 3, 3)

Hauteur

Largeur

channels

=



(4, 4, 1)

Convolution

4	5	8	7
1	8	8	8
3	6	6	4
6	5	7	8

4 x 4

*

1	4	1
1	4	3
3	3	1

3 x 3

=

122	148
126	134

2 x 2

Convolutional Matrix

1	4	1
1	4	3
3	3	1

3 x 3

=

1	4	1	0	1	4	3	0	3	3	1	0	0	0	0	0
0	1	4	1	0	1	4	3	0	3	3	1	0	0	0	0
0	0	0	0	1	4	1	0	1	4	3	0	3	3	1	0
0	0	0	0	0	1	4	1	0	1	4	3	0	3	3	1

4 x 16

4	5	8	7
1	8	8	8
3	6	6	4
6	5	7	8

4 x 4

=

4
5
8
7
1
8
8
8
3
6
6
4
6
5
7
8

16 x 1

Convolutional Matrix

1	4	1	0	1	4	3	0	3	3	1	0	0	0	0	0
0	1	4	1	0	1	4	3	0	3	3	1	0	0	0	0
0	0	0	0	1	4	1	0	1	4	3	0	3	3	1	0
0	0	0	0	0	1	4	1	0	1	4	3	0	3	3	1

4 x 16

.

4
5
8
7
1
8
8
8
3
6
6
4
6
5
7
8

16 x 1

=

122
148
126
134

4 x 1

=

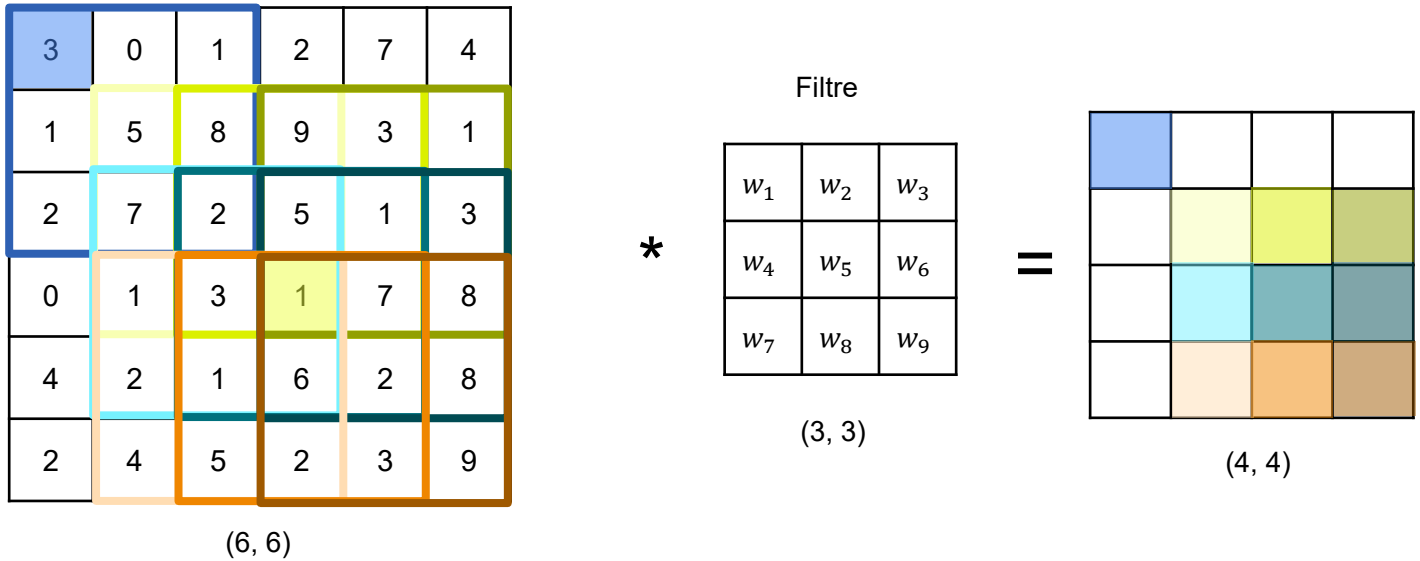
122	148
126	134

2 x 2

Le padding

Partie 7 : Computer vision

Sans padding



Avec padding

	3	0	1	2	7	4	
	1	5	8	9	3	1	
	2	7	2	5	1	3	
	0	1	3	1	7	8	
	4	2	1	6	2	8	
	2	4	5	2	3	9	

(6, 6) (8, 8)

Padding
p = 1

*

Filtre

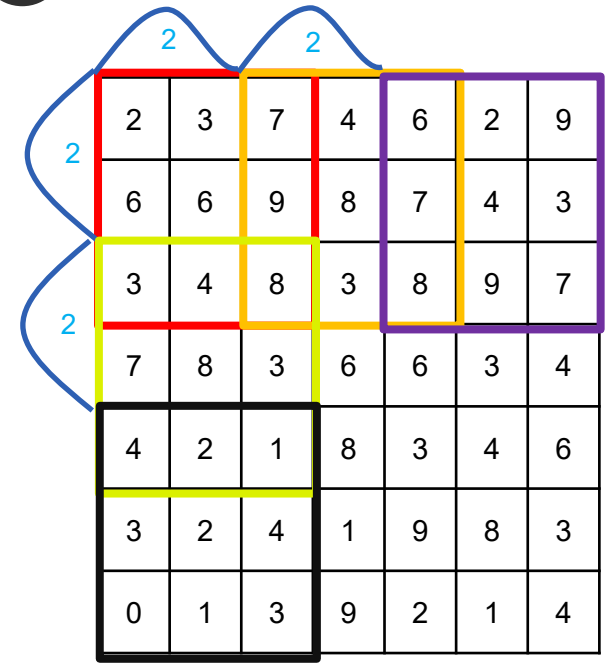
w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

(3, 3)

=

(6, 6)

Stride



(7, 7)

Stride
s = 2

filter

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

(3, 3)

*

=

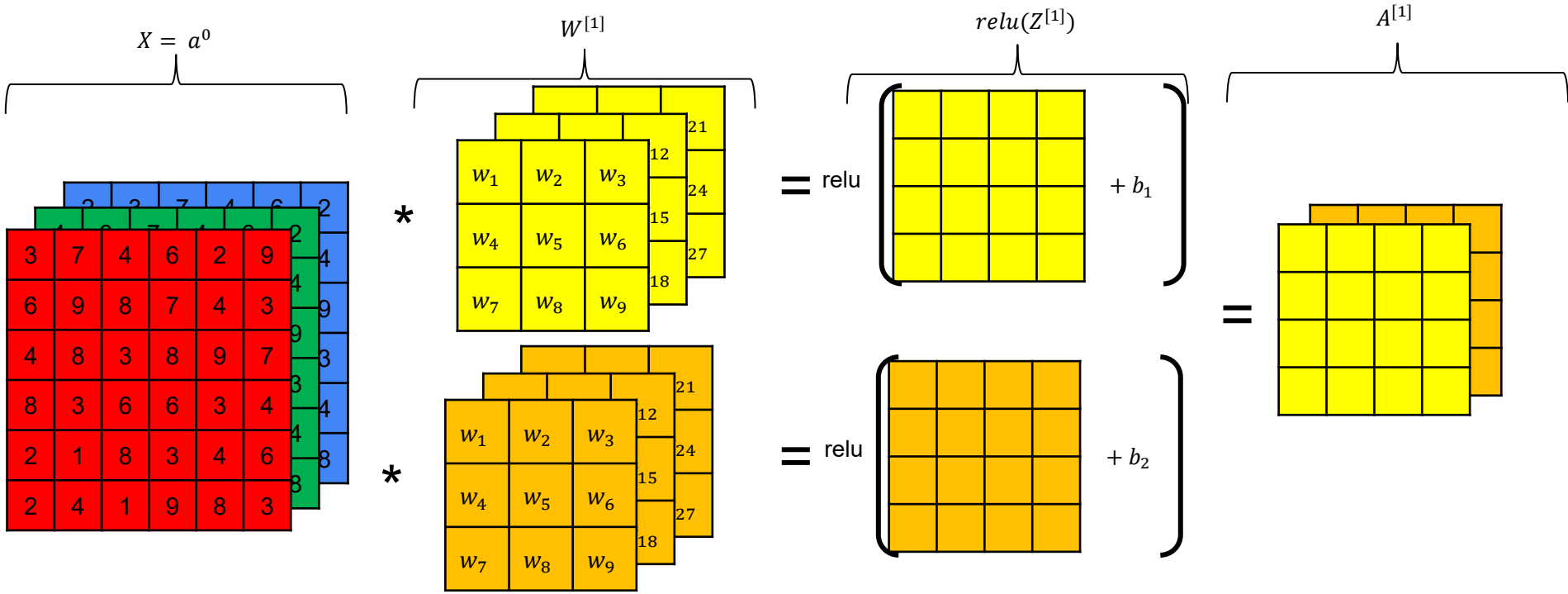
91	100	83
69	91	127
44	72	74

(3, 3)

Une couche de convolution

Partie 7 : Computer vision

Une couche de convolution



Le pooling

Partie 7 : Computer vision

Maintenir de l'invariance

Entrée

0	0	0	0
0	0	0	0
0	0	1	0
0	0	0	1

*

Fenêtre

1	0
0	1

=

Sortie

0	0	0
0	1	0
0	0	2

Max = 2

Ne change pas

Max = 2

Entrée

1	0	0	0
0	1	0	0
0	0	0	0
0	0	0	0

*

Fenêtre

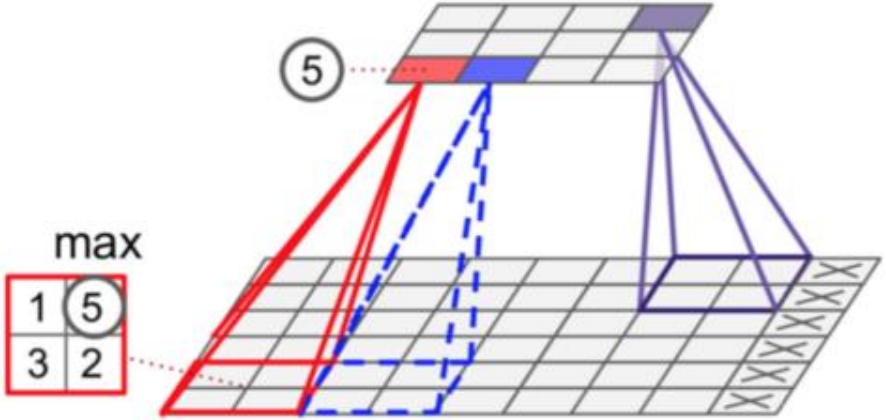
1	0
0	1

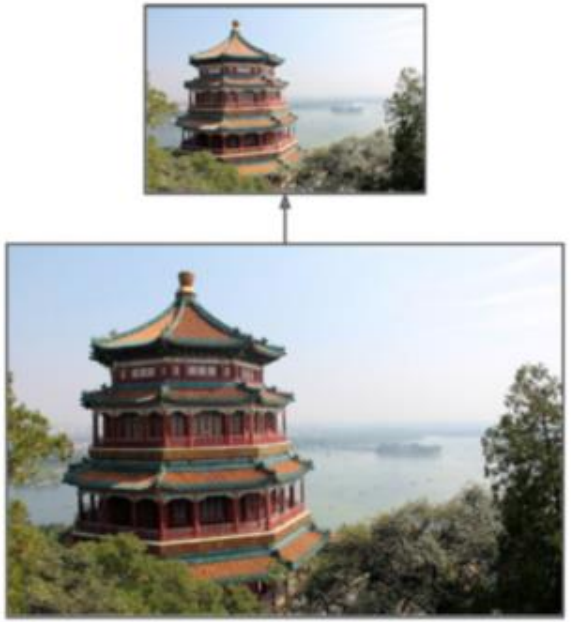
=

Sortie

2	0	0
0	1	0
0	0	0

Application sur une image





Pooling layer: Max pooling

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

(6, 6)

Hyperparameters:

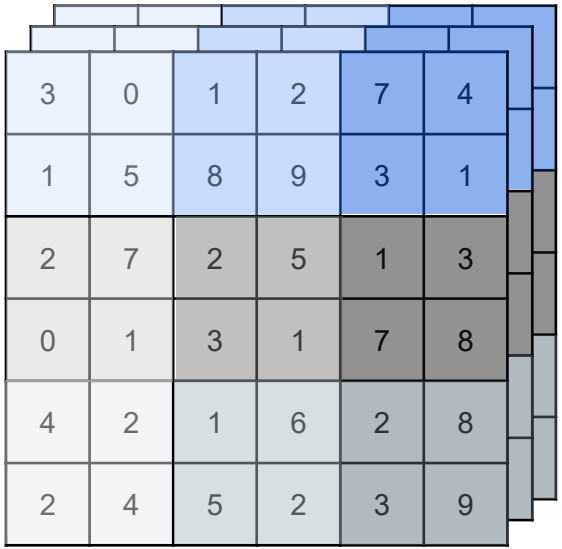
$f = 2$

$s = 2$

5	9	7
7	5	8
4	6	9

(3, 3)

Pooling layer: Max pooling

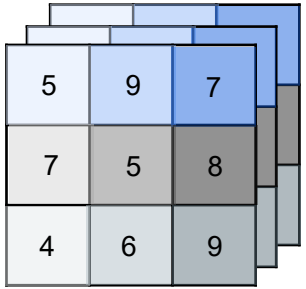


(6, 6)

Hyperparameters:

$f = 2$

$s = 2$



(3, 3)

Pooling layer: Average pooling

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

(6, 6)

Hyperparameters:

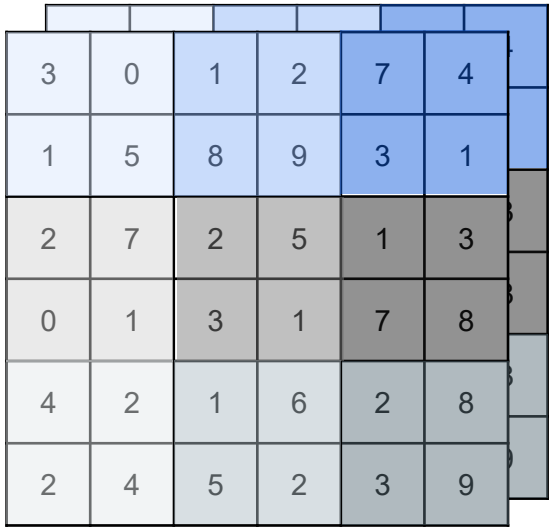
$f = 2$

$s = 2$

2.25	5	3.75
2.5	2.75	4.75
3	3.5	5.5

(3, 3)

Pooling layer: Average pooling

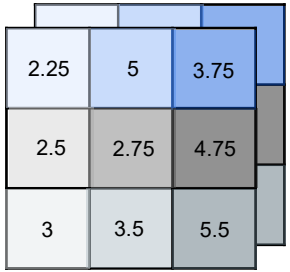


(6, 6, 2)

Hyperparameters:

$f = 2$

$s = 2$



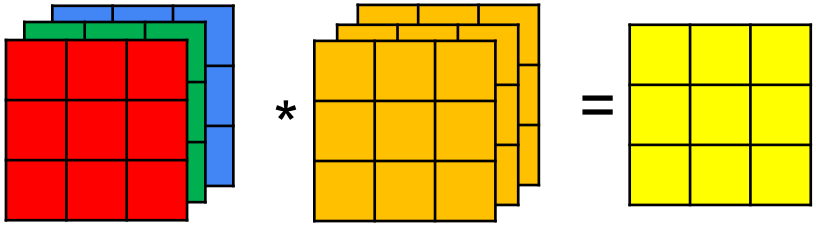
(3, 3, 2)

Les trois blocs du CNN

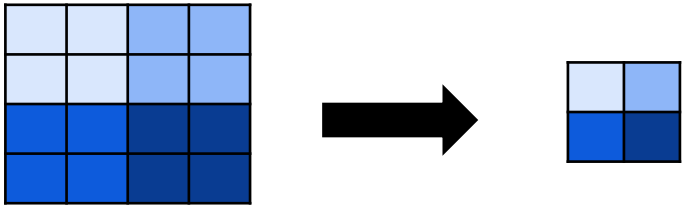
Partie 7 : Computer vision

Les trois blocs du CNN

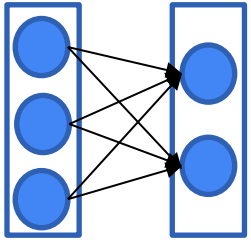
Convolutional bloc



Pooling bloc

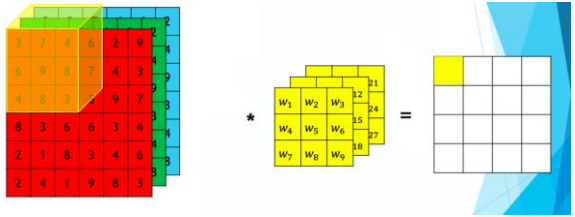


Fully connected bloc

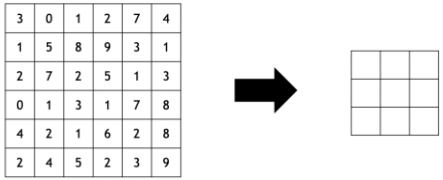


Les trois blocs du CNN

Convolution

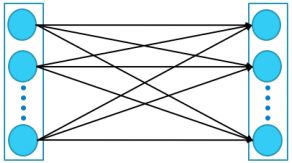


Pooling

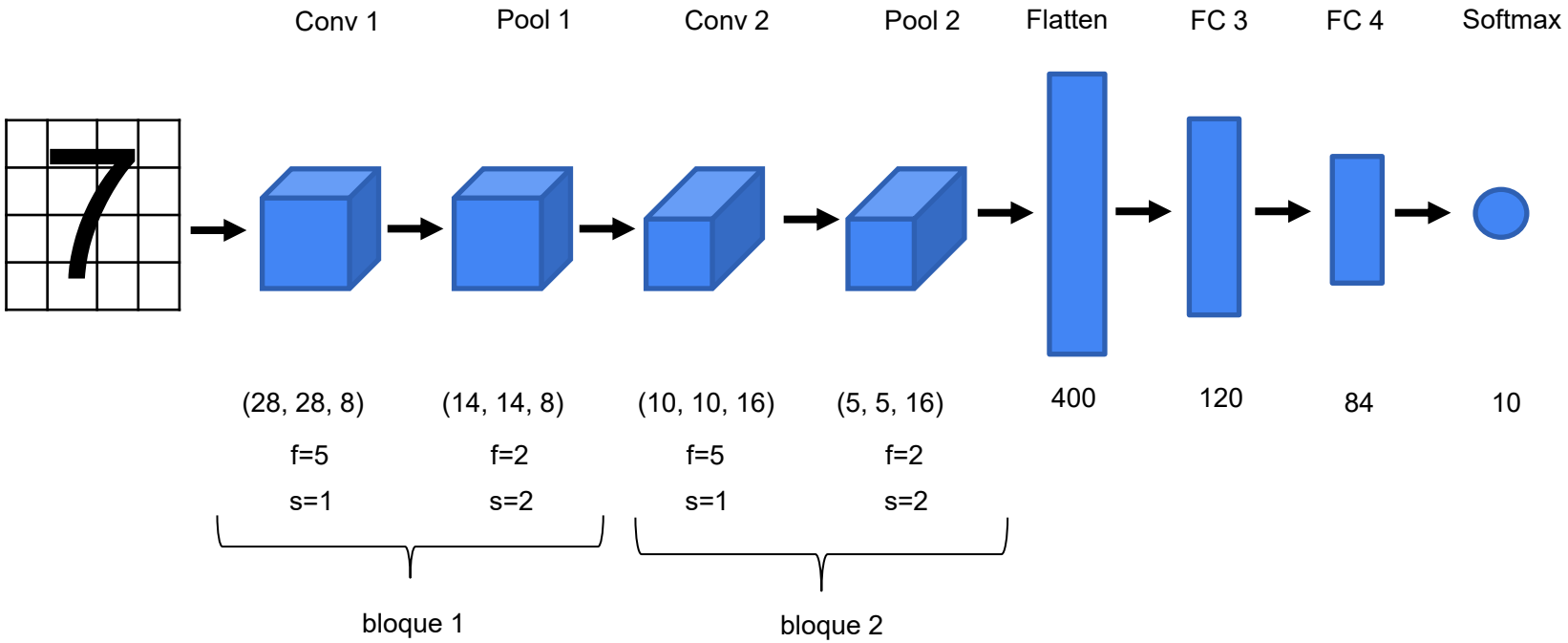


Flatten

Couche dense



LeNet-5 $\approx 60k$ *paramètres*



LeNet-5

	Activation shape	Activation size	# parameters
Input:	(32, 32, 3)	3 072	0
CONV1 (f=5, s=1)	(28,28, 8)	6 272	208
POOL1	(14, 14, 8)	1 568	0
CONV2 (f=5, s=1)	(10, 10, 16)	1 600	416
POOL2	(5, 5, 16)	400	0
FC3	(120, 1)	120	48 001
FC4	(84, 1)	84	10 081
Softmax	(10, 1)	10	841